When Events Speak: MLLM-Guided Video Retrieval with Temporal Reranking

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Abstract. The rapid expansion of online video content has intensified the need for efficient retrieval of specific moments. However, most existing video retrieval methods fail to capture short-lived events that occur within only a few frames and overlook dynamic cues such as motion, audio, and dialogue. This lack of event awareness prevents the system from understanding how actions evolve, leading to fragmented interpretations of event sequences. To address these challenges, we propose an event-aware video retrieval system that integrates MLLM-based event capturing with a temporal reranking mechanism. The MLLM captures both coarse-grained and fine-grained events, providing semantically rich and temporally aligned descriptions. The temporal reranking module then re-assesses the entire event sequence, dynamically relinking results based on recomputed sequence scores to maintain temporal coherence. This design allows the system to retrieve results that are both semantically meaningful and temporally coherent. Evaluated on the Ho Chi Minh AI Challenge 2025 under the team name LunchRetrieval, our system achieved 85 out of 88 correct results with accuracy above ninety-five percent across three evaluation stages, demonstrating strong robustness in complex event-sequence retrieval.

Keywords: Multimodal retrieval system · Event-aware video retrieval · Temporal reranking · Event filtering · Event capturing · Semantic video understanding

1 Introduction

With the rapid growth of video content across platforms such as social media, education, and entertainment, video retrieval has become increasingly essential. Every day, vast amounts of video data are created-YouTube alone sees over 500 hours of video uploaded per minute-posing significant challenges in efficiently locating specific information or moments within this massive data stream. Video retrieval tasks aim to identify and extract particular elements from videos, such

as scenes, spoken words, or objects, to support diverse applications ranging from media search to knowledge discovery.

Over the years, several international competitions have been organized to benchmark the performance of retrieval systems. Most of these challenges, such as the Lifelog Search Challenge (LSC) [1] and the Video Browser Showdown (VBS) [16], primarily focus on two fundamental tasks: Known-Item Search (KIS) and Visual Question Answering (VQA). These tasks evaluate a system's ability to locate specific video segments from textual or visual cues and to reason about content within individual frames. Recently, the Ho Chi Minh AI Challenge (HCMAI) 2025 [4] introduced a new task - Temporal Retrieval and Alignment of Key Events (TRAKE)-which emphasizes retrieving videos based on *semantic event sequences* and aligning each event with its corresponding keyframe. This addition highlights a crucial capability still missing in most retrieval systems: understanding and linking events coherently over time.

Despite advances in video retrieval systems such as SnapSeek [7] and NII-UIT [6], existing approaches still struggle to model complex event dynamics. They often fail at temporal reasoning, missing coherent links between related actions over time. For example, Fusionista [8] may detect a person opening a car door and starting the engine but cannot infer that these belong to one continuous event. Moreover, many methods rely on sparse keyframes extracted with models like CLIP [11] or CLIP2Video [5], which, while efficient, disrupt temporal continuity and overlook short but meaningful moments. Consequently, current systems often lose contextual coherence and fail to capture subtle transitions essential for accurate temporal understanding.

To overcome these challenges, we introduce an **Event Capturing and Filtering** module at the core of our multimodal video retrieval system, specifically designed to resolve failures in semantic and temporal understanding. Using a sliding-window mechanism and a Multimodal Large Language Model (MLLM) [18], the module moves beyond raw keyframes to generate detailed captions that describe key events. This process effectively captures short-lived actions and subtle semantic cues, addressing the limitations of conventional frame-based approaches. Subsequently, to ensure logical consistency across event sequences, our **Temporal Reranking** module dynamically recomputes and relinks retrieved events instead of merely aggregating similarity scores. Each new query triggers a re-evaluation of previous results, adjusting their temporal relationships to preserve global sequence coherence. This process penalizes incorrect ordering and long time gaps, ensuring the retrieved chain forms a consistent narrative. To enrich contextual understanding, the system also integrates Optical Character Recognition (OCR) and Automatic Speech Recognition (ASR) for textual and auditory cues. Through this integration of event-aware capturing, adaptive temporal relinking, and multimodal fusion, the system achieves precise and temporally consistent video retrieval.

Our system, developed under the team name **LunchRetrieval**, was evaluated in the Ho Chi Minh City AI Challenge 2025 qualification round, achieving **85 out of 88 correct results** across all tasks, including KIS, VQA, and

TRAKE, with an overall accuracy **exceeding ninety-five percent** through three evaluation stages. The system delivered an **outstanding overall performance** in the final round, demonstrating the effectiveness of our event-aware temporal reranking and multimodal filtering pipeline in enhancing retrieval precision and consistency for complex video queries.

2 Related Work

Recent advancements in video retrieval systems have significantly enhanced efficiency and accuracy, particularly through the integration of multimodal approaches that combine various features such as objects, visible text, colors, and speech. Leveraging these features from video frames and sequences, modern systems excel in handling diverse search tasks with precision. Competitions like the VBS and LSC highlight the effectiveness of combining search techniques, such as image-to-image retrieval, color detection, and text-based queries. The fusion of these methods, along with the use of image generation and multimodal algorithms, enables systems to tackle complex search scenarios and deliver high-accuracy results.

2.1 Temporal Search

Temporal Search is also a widely used approach that has been applied in many competitions, such as the VBS and HCMAI. For instance, in Fusionista [8] from VBS 2025, the connection between a sequence of queries and a video is typically measured using a matching score:

$$\text{Score}(Q, V) = \max_{(i_0, i_1, \ldots, i_k) \in \mathcal{I}} (\text{sim}(Q_0, V_{i_0}) + \cdots + \text{sim}(Q_k, V_{i_k})) \tag{1}$$

- Q is a the query sequence and V is a sequence of video frames.
- $\mathcal{I}$ is the set of all index combinations $(i_0, i_1, \ldots, i_k)$ such that $1 \leq i_0 < i_1 < \cdots < i_k \leq n$, ensuring the correct temporal order.
- $\text{sim}(Q_j, V_{i_j})$ represents the similarity between sub-query Q_j and frame V_{i_j}, which can be computed using cosine similarity, CLIP score, or other similarity measures.

The selected frame sequence is the one with the highest total matching score according to Equation (1)

Other teams have also extended the temporal search paradigm. The PraK Tool [12] introduces Bayesian relevance feedback, using probabilistic user feedback to iteratively refine retrieval scores and focus on segments most likely to be relevant. Similarly, the NII-UIT system [6] employs a Dynamic Temporal Search that first expands the temporal range of top-ranked results, then restricts subsequent searches to these regions. This multi-stage refinement preserves temporal continuity while narrowing the search space for more accurate and efficient event retrieval.

While prior systems such as Fusionista and NII-UIT introduced temporal search mechanisms, their methods remain largely local, evaluating queries through additive similarity scores or restricted temporal windows, resulting in fragmented or misaligned event sequences. Likewise, probabilistic feedback approaches like the PraK Tool improve relevance through user interaction but lack structural modeling of event progression. These limitations highlight the need for a retrieval framework capable of reasoning over full event chains and maintaining global temporal coherence, which motivates the design of our proposed approach.

2.2 Video Events Understanding

HORUS [10] employs MLLMs to extract multimodal features and expand user queries, enabling natural-language search over frame and segment embeddings. While effective for semantic matching, its retrieval remains largely frame/segment-centric and does not explicitly model how events relate across time, which can miss short-lived actions and cross-event dependencies.

VISIONE [2], the leading team in VBS 2024, enhances retrieval robustness through late fusion of multiple strong cross-modal encoders, OpenCLIP ViT-L/14, CLIP2Video [5], and ALADIN [9]. This design achieves high frame-level matching accuracy across textual and visual modalities. However, its fusion strategy remains largely similarity-driven, focusing on isolated matches rather than preserving the logical flow between temporally related events.

NII-UIT, the overall winner of VBS 2025, extends this idea by integrating multiple query types (text, image, query expansion, and object filtering) and combining normalized module scores into a final rank. It introduces a *Dynamic Temporal Search* module that expands temporal windows around top-ranked results to refine subsequent searches. While this mechanism improves local continuity, it still relies on sparse keyframes, which can miss short-lived actions and struggle to maintain long-range event consistency.

Across these systems, the common pipeline prunes visually similar keyframes to save memory and latency. This thinning breaks continuity, privileges isolated matches, and overlooks dynamic cues (motion, audio, dialogue), reducing accuracy whenever a query depends on how actions unfold over time.

To overcome these limitations, we propose a unified event-aware framework that integrates temporal reasoning and multimodal understanding, enabling more coherent and semantically accurate retrieval.

3 WESp (When Events Speak) System and Event-Aware Reranking Temporal Search Approach

WESp is an advanced multimodal retrieval framework designed to deliver precise and contextually relevant information from large-scale video archives. It unifies multiple retrieval strategies, textual, visual, OCR, ASR, and object-tag

based within a single architecture, enabling the system to capture both semantic meaning and perceptual similarity. At its core, WESp introduces two key innovations: an **Event Filtering module** that isolates semantically meaningful moments, and a **Temporal Reranking mechanism** that realigns results into coherent, time-consistent sequences. Together, these components transform fragmented retrieval outputs into logically structured narratives. The front-end provides an intuitive dual-panel interface for interactive query refinement and result exploration, empowering users to trace related frames, events, and timelines seamlessly. By integrating deep multimodal understanding with adaptive temporal reasoning, WESp achieves accurate, coherent, and user-oriented video retrieval across diverse search tasks. System overall pipeline is illustrated in Fig 1

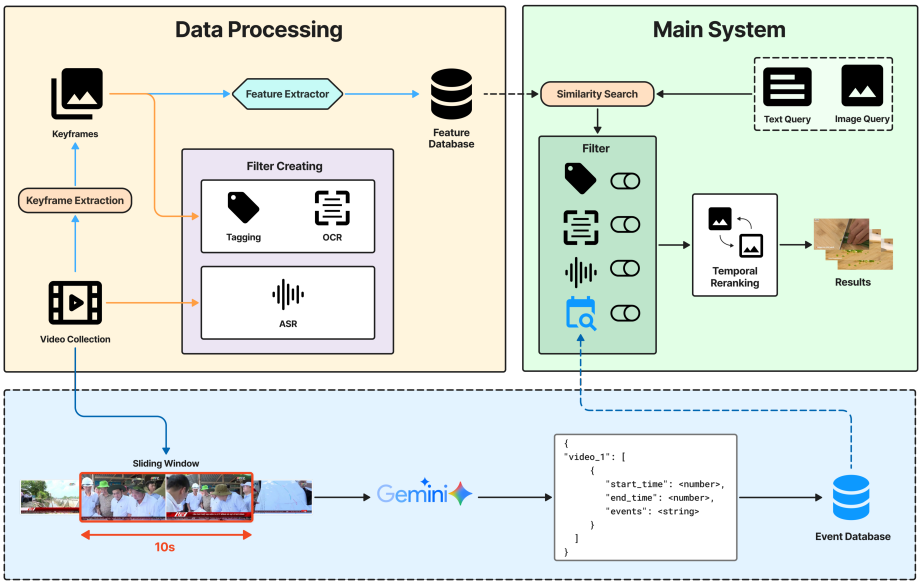


Fig. 1: System pipeline showing multimodal filtering and event-based retrieval. The filter module integrates Tagging, OCR, ASR, and Event Capturing to generate temporal constraints that refine similarity search and ensure coherent, context-aware results.

3.1 Similarity Search

Our proposed system leverages the advanced capabilities of the SigLIP2 [15] vision-language models, utilizing a multi-scale strategy with its large, giant, and so400m variants. This approach allows us to dynamically select the optimal "scope" for any query, capitalizing on SigLIP2's powerful core features, including its rapid encoding speeds, and deep contextual understanding of objects. Furthermore, it supports flexible, mixed-language queries to enhance user ex-

perience. To further improve robustness, we also incorporate the BEiT-3 [17] model, valued for its complementary strength in image-text alignment. The final results are generated using a rank fusion algorithm that combines the outputs from both the SigLIP2 variants and BEiT-3, ensuring a more comprehensive and accurate retrieval by leveraging the unique strengths of each model.

3.2 Multimodal Understanding and Filtering

To enable fine-grained understanding across text, speech, and visual content, our system integrates three complementary modules - Optical Character Recognition , Automatic Speech Recognition, and Object Tagging - within a unified multimodal filtering framework.

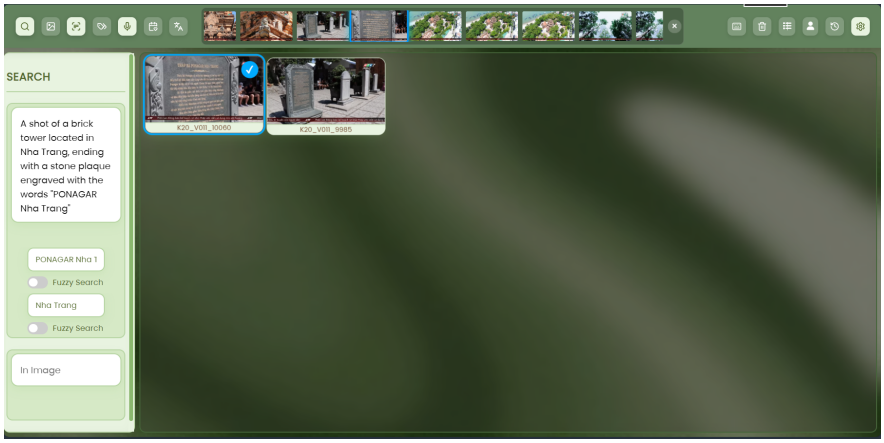


Fig. 2: Overview of the multimodal filtering process combining OCR, ASR, and object-based filters to reduce the search space and improve retrieval efficiency.

For OCR, we employ *Gemini-2.0-Flash* [14], a multimodal large language model capable of recognizing scene text under challenging conditions such as blur or partial occlusion. It can also reconstruct incomplete words, improving robustness on dynamic or low-quality frames. For ASR, we utilize *Gemini-2.5-Pro* [3], which exhibits strong zero-shot and multilingual performance. Six-second audio windows around each keyframe are transcribed to align spoken content with visual scenes. For object tagging, we integrate *RAM++* [19], state-of-the-art (SOTA) object and scene recognition model that produces semantically rich tags capturing both objects and contextual relationships.

Rather than aggregating scores across modalities, these modules serve as **modality-specific filters** to progressively narrow the search space. Each filter removes frames that fail to meet corresponding textual, auditory, or visual constraints, while a **filter fusion** mechanism allows multiple filters to be combined

with fuzzy matching for noisy inputs. As illustrated in Fig. 2, this multimodal filtering process reduces noise and computation overhead, resulting in faster and more accurate retrieval across diverse query types.

3.3 Event Capturing and Filtering With MLLM

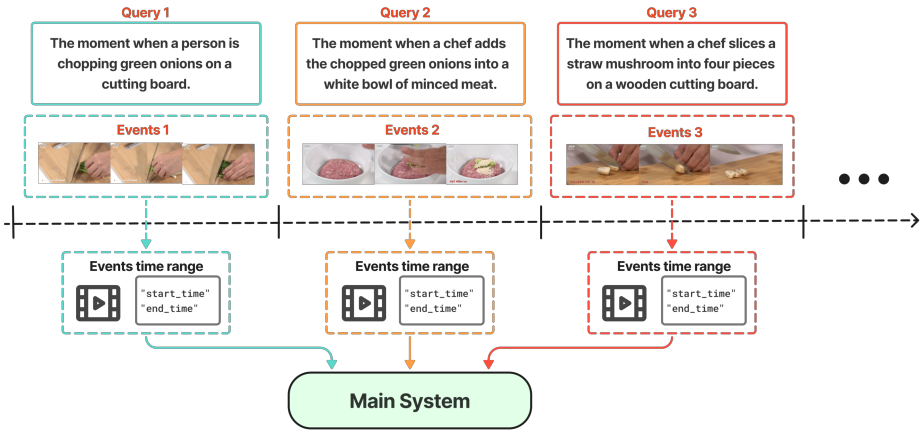


Fig. 3: **Progressive Event Filtering Mechanism.** Each query (*Q1, Q2, Q3*) retrieves the top-matched events from the video, represented as 10-second segments with 2-second overlap. The corresponding temporal intervals (`start_time`, `end_time`) are then converted into constraints that define valid search regions. When subsequent queries are added, the filter dynamically updates to include only events that are both semantically relevant and temporally consecutive to previous results. These refined intervals are passed to the main retrieval system, ensuring coherent and temporally consistent event linking.

To extract meaningful events from videos, we design an **Event Capturing** module based on a sliding-window mechanism. Each video is divided into continuous 10-second segments with a 2-second overlap between adjacent windows to ensure temporal continuity and prevent boundary actions from being missed. The 10-second duration allows the model to capture complete semantic events while keeping each segment short enough to maintain focus on a single context.

Unlike conventional systems where keyframes serve solely as static visual representations for search, in our framework they act as *anchors* representing short temporal clips. Each segment is processed as a coherent mini-clip rather than an isolated frame, allowing the model to perceive motion continuity and subtle contextual transitions within that short time span. This enables the system to recover short-lived or transient actions that typical keyframe-based pipelines fail to capture.

Each segment is then analyzed by the *Gemini-2.0-Flash* Multimodal Large Language Model (MLLM), which jointly interprets both visual and auditory signals to generate concise event descriptions. The MLLM plays a central role in capturing non-textual dynamic cues such as movement, gesture, and object interactions that cannot be inferred from ASR or OCR alone. While ASR and OCR modules contribute essential linguistic and textual information, they serve as complementary inputs that enrich the MLLM's understanding rather than substitutes for it. By combining these sources, the MLLM produces semantically rich, temporally aligned event captions that reflect not only what is seen or heard, but how actions evolve through time.

After the event descriptions are generated, the **Event Filtering module** refines the retrieval process by selecting only the temporally and semantically relevant parts of the video. Given a user query, the system encodes both the query and the event captions into a shared embedding space using the same multimodal encoder, ensuring consistent cross-modal alignment. A similarity score is computed between the query embedding and each event representation, and the top-k most relevant events are selected as candidate segments.

To construct the filter, the temporal intervals of the top-ranked events are converted into constraints that limit the frame-level search space. Only frames within these time ranges are preserved for further retrieval, allowing the system to focus exclusively on semantically rich segments. When subsequent queries (e.g., the second or third in a temporal sequence) are introduced, the filter dynamically adapts, selecting only those events that not only match the new query but also occur shortly after the previously retrieved events. This ensures stronger temporal linkage and maintains a coherent progression of actions. As illustrated in **Fig. 3**, the search space progressively narrows as each new query is added, allowing the system to concentrate computation on temporally relevant regions while discarding unrelated segments.

This hierarchical and adaptive filtering design makes the module highly robust to noise by discarding visually repetitive or contextually irrelevant frames while maintaining meaningful temporal dependencies. Furthermore, the event filter can seamlessly integrate with other modalities such as OCR or ASR filtering to reinforce semantic consistency when textual or spoken cues are present.

Through this mechanism, the **Event Filtering module** efficiently narrows the search to contextually relevant intervals, preserves temporal coherence across queries, and significantly improves retrieval accuracy in long, multi-event video sequences.

3.4 Temporal Reranking and Sequence Relinking

To ensure logical consistency across event-based queries, we introduce a **Temporal Reranking Mechanism** that dynamically recomputes and relinks retrieved results across the entire query chain. Unlike traditional *beam search* [8], which simply aggregates similarity scores and selects the best local sequence, our method iteratively refines the temporal relationships between all events, maintaining global consistency across the evolving query chain.



Fig. 4: Improve KIS performance by enforcing temporal coherence across event sequences

Core Concept. When a new query is introduced, the algorithm does not treat it as an isolated retrieval. Instead, it re-evaluates the existing chain of results by recalculating temporal compatibility between past and current events within each video. This involves updating not only the local pairwise temporal scores but also the cross-event links that connect earlier results to the newly retrieved ones, ensuring that the sequence remains temporally plausible.

Reranking & Relinking (Formalization). For neighbor queries $X \to Y$ within the same video, let the hit sets be (t_i^X, s_i^X) and (t_j^Y, s_j^Y). Define the *temporal distance* and *blended similarity* as:

$$D_{ij}^{XY} = |t_i^X - t_j^Y|, \qquad B_{ij}^{XY} = w_A s_i^X + (1 - w_A) s_j^Y.$$

A distance penalty $\phi(d)$ is applied depending on the selected mode:

$$\phi_{\exp}(d) = 1 - e^{-\gamma d/\alpha}, \qquad \phi_{\text{sqrt}}(d) = \min\left(\sqrt{1 + \left(\frac{\beta d}{\alpha}\right)^2} - 1,\ 1\right).$$

The temporal score matrix is then given by:

$$\boxed{M_{ij}^{XY} = B_{ij}^{XY}\left(1 - \lambda\,\phi(D_{ij}^{XY})\right)} \qquad M_{ij}^{XY} = 0 \text{ if } D_{ij}^{XY} \geq T_{\max}. \tag{1}$$

,

First link ($A \to B$). For each A_i, select the best matching connection:

$$m_i^{AB} = \max_j M_{ij}^{AB}, \qquad j_i^* = \arg\max_j M_{ij}^{AB}.$$

Update the confidence score and record the link:

$$\boxed{\tilde{s}_i^A = \tfrac{1}{2}\big(s_i^A + m_i^{AB}\big), \qquad \text{chain}_i(B) \leftarrow j_i^*.} \tag{2}$$

If no valid match exists for B, apply score decay:

$$\tilde{s}_i^A \leftarrow \eta\, \tilde{s}_i^A, \qquad \eta = 0.1.$$

,

Subsequent links ($B \to C$). Find the strongest bridge:

$$(u^*, v^*) = \arg\max_{u,v} M_{uv}^{BC}, \qquad L^{BC} = \max_{u,v} M_{uv}^{BC}.$$

Then for each A_i, recompute the link between A and B^*, and blend it with the bridge strength:

$$r_i^{A\to B^*} = \big[w_A \tilde{s}_i^A + (1 - w_A)\tilde{s}_{u^*}^B\big]\Big(1 - \lambda\, \phi\big(|t_i^A - t_{u^*}^B|\big)\Big),$$

$$\boxed{\tilde{s}_i^A \leftarrow \tfrac{1}{3}\big(\tilde{s}_i^A + r_i^{A\to B^*} + L^{BC}\big), \quad \text{chain}_i(B) \leftarrow u^*, \quad \text{chain}_i(C) \leftarrow v^*.} \tag{3}$$

If no valid neighbors exist, apply decay:

$$\tilde{s}_i^A \leftarrow \eta\, \tilde{s}_i^A.$$

Eqs. (1)-(3) implement vectorized gap penalties, first-link averaging, and *global* sequence relinking: each new query triggers chain-wide score updates and link rewiring, yielding a temporally coherent timeline rather than a locally optimal (beam-like) path.

Example. For the sequence "*open door → walk out → drive away*", later cues (e.g., "drive away") can improve earlier matches by revising $A \leftrightarrow B$ links via Eq. (3), producing a smooth, time-aligned narrative.

As shown in **Fig. 4**, this chain-level reranking substantially improves the ordering of retrieved shots, ensuring that the final output reflects a coherent and contextually accurate event sequence.

3.5 WESp User Interface

Our system features an intuitive and collaborative user interface (UI) designed to optimize the efficiency of multi-modal video retrieval. Similar to conventional multi-modal systems, the interface supports multiple input modalities such as text, image, ASR, OCR, and object-based filtering. However, our system extends these capabilities with additional functional buttons (fig 5), including *Event Filtering* and *tag filtering*, enabling users to refine their queries with greater precision.

Beyond individual usage, the UI incorporates a Teamwork mode that facilitates seamless teamwork. This mode allows multiple users to interact with the system simultaneously, thereby improving communication and collective decision-making efficiency during the retrieval process.

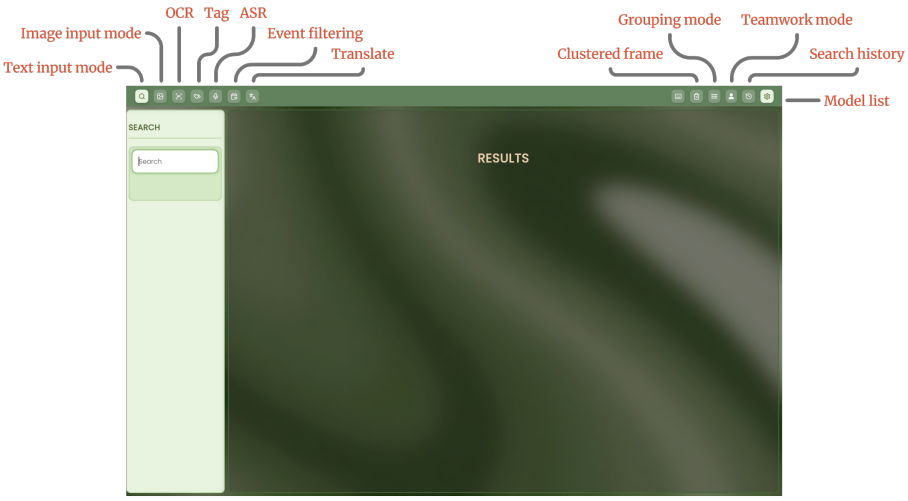


Fig. 5: User Interface - all features

To further enhance retrieval efficiency, the system integrates a clustering filtering module within the user interface. During large-scale video searches, redundant frames such as intros, transitions, or anchor shots often add noise and slow down retrieval. To address this, all keyframes are embedded into a visual feature space and clustered via K-Means based on similarity. Each cluster represents a consistent visual theme (e.g., studio scenes or logos), which users can inspect and remove interactively through the UI. Once a cluster is marked as irrelevant, all its frames are filtered out, reducing noise and accelerating retrieval by focusing on semantically meaningful content.

The interface provides two distinct visualization modes: Normal Mode and Grouping Mode. In Normal Mode, the system displays representative frames of the retrieved results, offering a concise and efficient browsing experience. Conversely, Grouping Mode presents all frames within each video, allowing users to examine the search results in more detail and better understand the contextual relationships among frames.

4 Experiments

4.1 Datasets

The dataset used in the Ho Chi Minh AI Challenge 2025 focuses on Vietnamese news and daily life videos, providing a diverse and representative collection for video analysis research. The dataset has a combined duration of approximately 300 hours, covering a wide range of social and news-related topics. To prepare the data, we applied the LMSKE [13] pipeline, using AutoShot [20] to segment videos into shots before selecting representative keyframes based on semantic changes. This preprocessing step efficiently reduced redundancy while preserving essential visual and contextual information. The final dataset contains around 581,000 keyframes, forming a robust benchmark for developing and evaluating advanced video retrieval systems.

4.2 Qualification Round Result

Our system was evaluated in the *Ho Chi Minh City AI Challenge 2025* qualification round, achieving **85 out of 88 correct answers** across all tasks, including KIS, VQA, and TRAKE, maintaining **accuracy exceeding ninety-five percent** through three evaluation stages. This strong performance demonstrates the effectiveness of our multimodal filtering, event-aware temporal reranking, and robust retrieval pipeline. Furthermore, the results highlight the system's ability to handle complex multimodal queries with high accuracy and strong temporal consistency, confirming its competitiveness among leading approaches in the challenge.

5 Discussion and Future Work

Our participation in the **Ho Chi Minh City AI Challenge 2025 Final Round** provided an opportunity to evaluate the full capabilities of the proposed system, **LunchRetrieval**, under real-world, time-constrained conditions. The final evaluation covered four main tasks: TKIS, VKIS, TRAKE, and VQA.

Across all tasks, our system delivered consistent and high-quality performance, achieving: **TKIS:** Outstanding, **VKIS:** Outstanding, **TRAKE:** Excellent, **VQA:** Excellent, **Overall Performance:** Outstanding,

These results confirm the effectiveness of the proposed event-aware retrieval pipeline combining **MLLM-based Event Capturing**, **Adaptive Event Filtering**, and **Temporal Reranking and Relinking** in retrieving complex, multi-event video sequences with strong semantic and temporal coherence. In particular, the reranking mechanism proved crucial for aligning sequential actions and reducing order inconsistencies across temporally dependent queries.

As illustrated in **Fig. 6**, our system successfully retrieved complete event chains, maintaining accurate temporal ordering and contextual relevance even in long and dynamic video scenes.

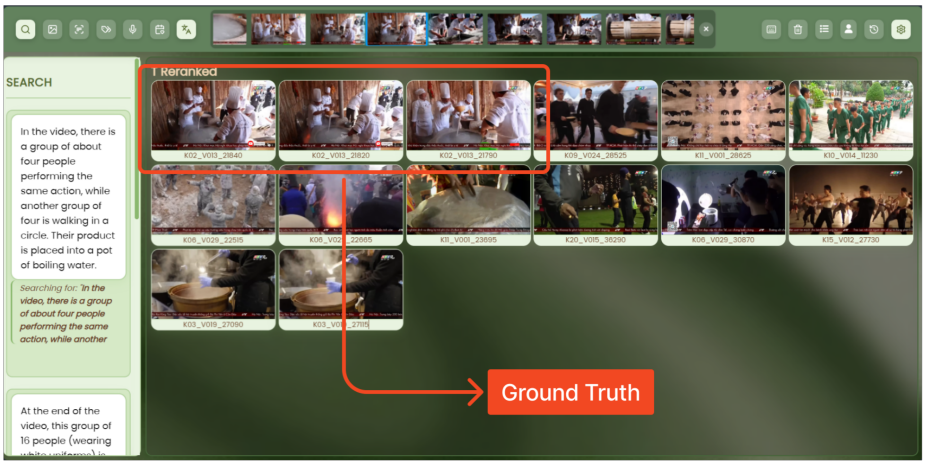


Fig. 6: **Final Round Results.** Example of LunchRetrieval's performance during the Ho Chi Minh City AI Challenge 2025 Final Round.

Despite this strong performance, certain limitations remain in handling overlapping or simultaneous events (e.g., "*speaking while driving*"), where temporal boundaries blur. In future work, we aim to integrate explicit motion tracking and fine-grained temporal fusion modules to better capture multi-threaded event structures, further enhancing the robustness of multimodal temporal reasoning.

6 Conclusion

In this paper, we introduced WESp, a multimodal video retrieval system that integrates multiple components, text, image, OCR, ASR, and object tagging, into a unified framework for precise and efficient information retrieval. The system further incorporates two key innovations: event-aware temporal reranking, which ensures logical consistency across sequential queries, and event filtering, which focuses the search on semantically relevant segments. These components collectively enhance the accuracy, coherence, and speed of retrieval in complex search scenarios. Evaluated on the Ho Chi Minh AI Challenge 2025 under the team name LunchRetrieval, our system successfully answered 85 out of 88 queries across KIS, VQA, and TRAKE tasks and achieving outstanding performance on the final round. The results underscore our system's proficiency in combining temporal reasoning and multimodal understanding for event-focused video retrieval.

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